

B. Tech Year IV Sem VII
A.Y. 2026-27 (MAY 2026 to September 2026)

Teaching Scheme:

Teaching Scheme (Hours)			Credits	Theory Marks			Tutorial/Practical Marks		Total Marks
Theory	Tutorial	Practical		ESE	IA	CSE	Viva	Term Work (TW)	
3	0	2	4	50	30	20	25	25	150

Included activities for assessment:

- Theory Exam
- Lab Experiments
- Assignments
- Quiz
- Viva
- Mini Project

Assessment Components and Weightage

- **Internal Assessment (IA)**
Total: 30 Marks Plan: 60 Marks
➤ Two Mid-Semester Examinations (MID-I and MID-II) of 30 marks each will be conducted during the semester, and the final Internal Assessment marks will be calculated based on the average performance of both examinations.
- **Continuous Semester Evaluation (CSE)**
Total: 20 Marks Plan: 120 Marks
➤ 7 Module-wise quizzes (7*10 = 70 Marks) will be conducted during the course, and 5 Assignments (5*10 = 50 Marks) will be given, and the performance will be considered for the 20 marks allocated to the CSE component.
- **Term work (TW)**
Total: 25 Marks
➤ Lab work carrying 15 marks will be evaluated along with post-lab exercises.
➤ Mini Project carrying 10 marks will be evaluated based on design, implementation, and demonstration.
- **End Semester Exam (ESE)**
Total: 50 Marks Plan: 60 Marks
➤ A 60-mark theory examination will be conducted for the End Semester Examination (ESE), out of which 50 marks will be counted for evaluation.
- **Viva**
Total: 25 Marks
➤ An oral viva examination will be conducted at the end of the course to evaluate the student's conceptual understanding and practical knowledge.

Relative grading will be applicable for assessment of the course. Failure in the subject will lead to remedial exam of all assessment components.

Students would need to attend at least 75% of the sessions to pass this course.

*Note: Any Type of Copying/Plagiarism in any component would result in an 'F' Grade.



Mr. Ajay Rupani
Subject Coordinator



Head of the department

Date: 9/6/2026

CSE, TW, and Partial Viva examinations will be conducted through the Google Classroom platform.

Course Evaluation

CO Attainment

Batch 2020-2024

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
01CT0718	CO1	3	2	1	0	3	0	0	0	0	0	0
	CO2	3	2	1	0	3	0	0	0	0	0	0
	CO3	3	2	1	0	3	0	0	0	0	0	0
	CO4	3	2	1	0	3	0	0	0	0	0	0
	CO5	3	2	2	0	3	0	0	0	0	0	0

Subject		Direct (80%)	Indirect (20%)	Total CO (100%)	Target	YES / NO
FPGA Based System Design (01CT0718)	CO1	62.82	15.00	77.82	54	YES
	CO2	45.65	14.50	60.15	54	YES
	CO3	55.77	15.00	70.77	54	YES
	CO4	47.30	16.00	63.30	54	YES
	CO5	47.30	15.00	62.30	54	YES

Batch 2021-2025

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
01CT0718	CO1	3	2	1	0	3	0	0	0	0	0	0
	CO2	3	2	1	0	3	0	0	0	0	0	0
	CO3	3	2	1	0	3	0	0	0	0	0	0
	CO4	3	2	1	0	3	0	0	0	0	0	0
	CO5	3	2	2	0	3	0	0	0	0	0	0

Subject		Direct (80%)	Indirect (20%)	Total CO (100%)	Target	YES / NO
FPGA Based System Design (01CT0718)	CO1	60.50	9.33	69.83	54	YES
	CO2	62.00	9.33	71.33	54	YES
	CO3	65.00	9.33	74.33	54	YES
	CO4	52.00	9.33	61.33	54	YES
	CO5	52.00	9.33	61.33	54	YES

Batch 2022-2026

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
01CT0718	CO1	3	2	1	0	3	0	0	0	0	0	0
	CO2	3	2	1	0	3	0	0	0	0	0	0
	CO3	3	2	1	0	3	0	0	0	0	0	0
	CO4	3	2	1	0	3	0	0	0	0	0	0
	CO5	3	2	2	0	3	0	0	0	0	0	0

Subject		Direct (80%)	Indirect (20%)	Total CO (100%)	Target	YES / NO
FPGA Based System Design (01CT0718)	CO1	37.17	20.00	57.17	60	NO
	CO2	40.67	16.00	56.67	60	NO
	CO3	62.00	16.00	78.00	60	YES
	CO4	77.00	20.00	97.00	60	YES
	CO5	68.00	20.00	88.00	60	YES

To set the current semester Course Outcome (CO) attainment targets, a simple and effective approach is to consider the CO-wise average attainment of the previous three batches and use this average as the reference target for the current semester.

Average CO Attainment Calculation (Last Three Years)

Overall Average Attainment

CO	Average Attainment	
CO1	68.27%	(77.82+69.83+57.17)/3
CO2	62.71%	(60.15+71.33+56.67)/3
CO3	74.36%	(70.77+74.33+78.00)/3
CO4	73.87%	(63.30+61.33+97.00)/3
CO5	70.54	(62.30+61.33+88.00)/3

The current semester CO attainment targets are set based on the CO-wise average attainment of the previous three academic batches to ensure a realistic and data-driven benchmark for continuous improvement.

PO Attainment

Batch 2020-2024

PO	PO THROUGH CIEs & SEE (80%)	PO THROUGH COURSE END SURVEY (20%)	Total (%)	Target (%)	YES / NO
PO1	51.77	15.10	66.87	60	YES
PO2	51.77	15.10	66.87	60	YES
PO3	51.02	15.08	66.10	60	NO
PO4			0.00	60	NO
PO5	51.77	15.10	66.87	60	YES
PO6			0.00 ;	60	NO
PO7			0.00	60	NO
PO8			0.00	60	NO
PO9			0.00	60	NO
PO10			0.00	60	NO
PO11			0.00	60	NO
PO12			0.00	60	NO
PSO1	51.77	51.77	51.77	60	NO
PSO2			0.00	60	NO

Batch 2021-2025

PO	PO THROUGH CIEs & SEE(80%)	PO THROUGH COURSE END SURVEY(20%)	Total (%)	Target (%)	YES / NO
PO1	58.30	9.33	67.63	60	YES
PO2	58.30	9.33	67.63	60	YES
PO3	57.25	9.33	66.58	60	YES
PO4			0.00	60	NO
PO5	58.30	9.33	67.63	60	YES
PO6			0.00	60	NO
PO7			0.00	60	NO
PO8			0.00	60	NO
PO9			0.00	60	NO
PO10			0.00	60	NO
PO11			0.00	60	NO
PO12			0.00	60	NO
PSO1	58.30	58.30	58.30	60	NO
PSO2			0.00	60	NO

MARWADI UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
SUBJECT NAME: FPGA BASED SYSTEM DESIGN (01CT0718)

Batch 2022-2026

PO	PO THROUGH CIES & SEE (80%)	PO THROUGH COURSE END SURVEY (20%)	Total (%)	Target (%)	YES / NO
PO1	56.97	18.40	75.37	60	YES
PO2	56.97	18.40	75.37	60	YES
PO3	58.81	18.67	77.47	60	YES
PO4			0.00	60	NO
PO5	56.97	18.40	75.37	60	YES
PO6			0.00	60	NO
PO7			0.00	60	NO
PO8			0.00	60	NO
PO9			0.00	60	NO
PO10			0.00	60	NO
PO11			0.00	60	NO
PO12			0.00	60	NO
PSO1	56.97	56.97	56.97	60	NO
PSO2			0.00	60	NO

Summary

The CO attainment analysis of the subject *FPGA Based System Design (01CT0718)* over the last three academic batches indicates consistent academic performance and satisfactory achievement of learning outcomes. In Batch 2020–2024, all five COs successfully achieved the target attainment value of 54%. The overall average CO attainment for the batch was approximately 66.86%, with CO1 attaining the highest value of 77.82%, which reflects strong understanding of fundamental concepts of digital circuits among students. The students of this batch demonstrated better classroom participation and analytical ability, which positively contributed to higher direct attainment values.

Similarly, in Batch 2021–2025, all COs achieved the prescribed target of 54%, and the overall average CO attainment was approximately 67.63%. The attainment values remained stable and balanced across all COs, indicating the effectiveness of continuous internal assessment, tutorial sessions, assignments, and laboratory activities. The attainment was slightly higher compared to the previous batch, students were able to meet the expected learning outcomes through consistent academic monitoring and practical learning approaches. The batch showed moderate performance in analytical and application-based topics but maintained satisfactory overall attainment.

For Batch 2022–2026, the target attainment level was increased from 54% to 60% to encourage improved academic performance and higher learning standards. The batch achieved an overall average CO attainment of approximately 75.36%. CO3, CO4, and CO5 successfully achieved the revised target, while CO1 and CO2 attained 57.17% and 56.67%, respectively, slightly below the target value. This indicates that some students faced difficulty in understanding analytical and problem-solving-oriented concepts related to the course. The analysis also reveals that variations in students' foundational knowledge and academic preparedness influenced the attainment levels, particularly in technically challenging modules.

The three-year PO attainment analysis of the *FPGA Based System Design (01CT0718)* course reflects significant variation in attainment performance across different academic batches. In Batch 2020–2024, four Program Outcomes (POs), PO1, PO2, PO3, and PO5 successfully achieved the target attainment level of 60%, indicating effective course delivery, strong student participation, and good integration of theoretical and practical learning approaches. Only PSO1 did not achieve the target, suggesting limited domain-specific and research-oriented exposure during that cycle.

In Batch 2021–2025, a considerable increase in PO attainment was observed, where four Program Outcomes (POs), PO1, PO2, PO3, and PO5 achieved the target value, but PSO1 did not achieve the target. The improvement indicates that the implementation of practical learning activities, mini-projects, hands-on sessions, simulation-based exercises, peer-learning activities, and continuous assessment strategies positively contributed to the enhancement of student performance and outcome attainment.

For Batch 2022–2026, substantial improvement in PO attainment was observed. Four Program Outcomes (POs), PO1, PO2, PO3, and PO5 achieved the target value, but PSO1 did not achieve the target, which remained slightly below the target. The improvement indicates that the implementation of practical learning activities, mini-projects, hands-on sessions, simulation-based exercises, peer-learning activities, and continuous assessment strategies positively contributed to the enhancement of student performance and outcome attainment.

Based on the CO and PO attainment analysis, it is observed that students need improvement in conceptual understanding, analytical thinking, practical implementation, teamwork, and communication skills. To improve attainment in the current semester, the subject coordinator can conduct CO-based quizzes, application development based, simulation and hands-on laboratory activities. Additionally, continuous assessment, rubric-based evaluation, and application-oriented assignments will help strengthen problem-solving ability, practical exposure, and overall student performance, leading to better CO and PO attainment.

The combined analysis of 12th Science academic performance and CO attainment for the batches 2020–2024, 2021–2025, and 2022–2026 indicates that students' academic background has a direct influence on Course Outcome attainment in the subject *FPGA Based System Design (01CT0718)*. Batch 2022–2026 demonstrated the highest overall CO attainment, with all COs exceeding the target value and achieving an overall average attainment of approximately 75.36%. Batch 2021–2025 maintained stable and satisfactory performance with all COs achieving the prescribed target and an overall average attainment of around 67.63%. In Batch 2022–2026, the target was increased from 54% to 60%, and despite the higher target level, the batch achieved an overall average attainment of approximately 75.36%, where CO3, CO4, and CO5 met the revised target successfully, while CO1 and CO2 remained slightly below the target value. The analysis clearly shows that students with higher 12th Science percentages demonstrated better conceptual understanding, analytical ability, programming skills, and laboratory performance, which positively contributed to higher direct attainment and overall CO achievement.

The overall three-year average CO attainment is approximately 69.95%, which indicates satisfactory academic performance and effectiveness of the teaching-learning process. Students with strong academic foundations adapted more effectively to analytical and implementation-oriented topics, whereas students with moderate academic backgrounds required additional support through tutorial sessions, remedial lectures, and practical activities. Based on the analysis, it is recommended to maintain the CO attainment target at 70% for the current semester. To further improve attainment levels, activities such as remedial classes, unit-wise quizzes, peer learning, laboratory-oriented practice sessions, mini-project implementation, and continuous performance monitoring should be conducted regularly. These corrective measures will help strengthen conceptual understanding, improve problem-solving skills, and enhance overall student performance and CO attainment in future semesters.